Congestion-aware slotting

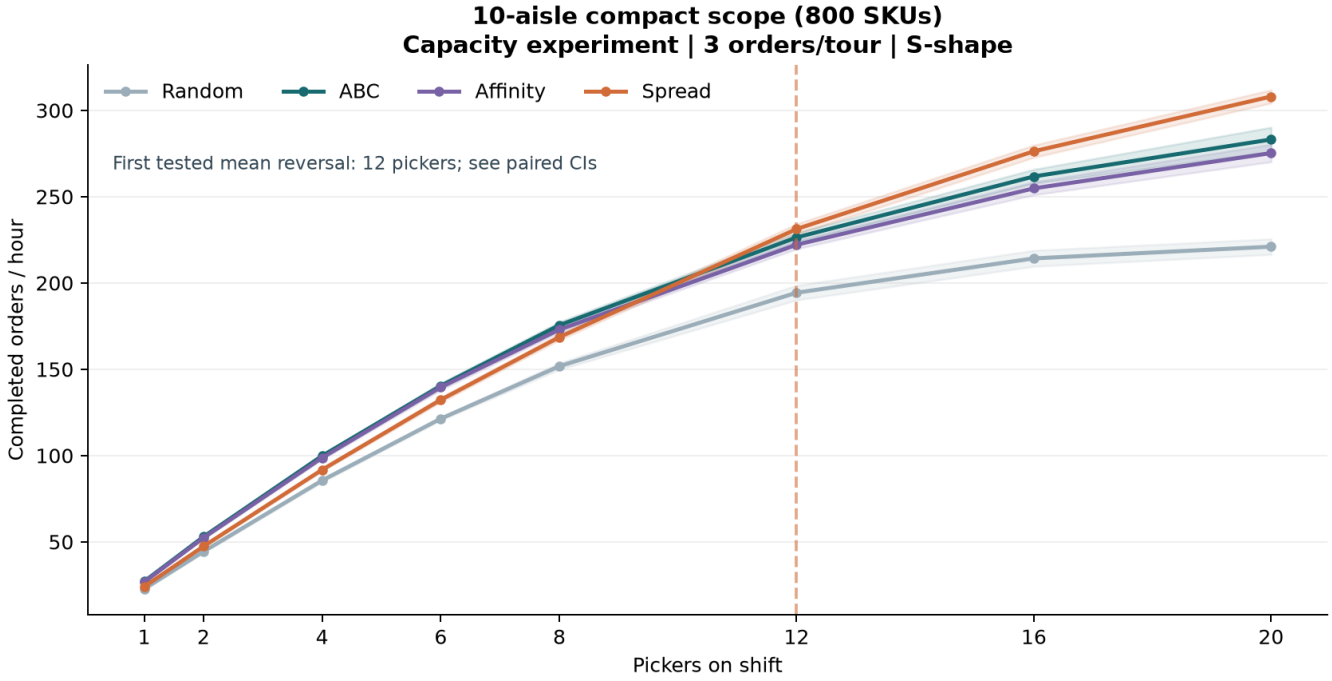
Concentrate first. Spread when staffing and operating rules justify it.

12 pickers
first tested mean crossover*

16 pickers
sustained simultaneous evidence*

183.2 h
paid labor / 1,000 orders**

*10-aisle compact model, k=3, no passing with entry control. **Larger-layout demand scenario; see page 2.



Bands: 95% t intervals across 10 replications. Model capacity, not observed warehouse output.

The result, and where it stops applying

At 20 compact-model pickers, Spread produces **308.0 vs. 283.1 orders/hour** for ABC (+8.8%), while blocking falls from **47.6% to 35.7%** of active time. It pays for 16.1% more single-order travel than ABC. The first positive tested paired mean is at 12 pickers; the simultaneous interval includes zero there and excludes zero from 16 onward. Do not report the interpolated 10.34 as a precise threshold.

The larger, 115-aisle layout has **no crossover through 20 total pickers**. Extending staffing to comparable density produces the first tested reversal at **138 pickers (12 per ten aisles)**, following an ABC advantage at 92. Density, not an unqualified total headcount, is the useful comparison.

Critical scope limit: top-800 storage covers 50.3% of held-out lines but only 8.0% of complete orders. The 9,149-SKU design covers 90.0% of lines and 53.0% of complete orders. Orders touching excluded items are excluded, not shortened. These curves do not price a complete warehouse with a long-tail process.

Operating recommendation: optimize batching first

At a labeled **33.1 eligible orders/hour** in the larger layout, select **ABC, k=5, six pickers**. Choose the smallest tested staffing whose lower 95% capacity bound exceeds demand / 0.85. There is no observed facility demand or wage; $25/hour is illustrative.

Larger-layout option	Pickers	Paid h/1,000	Oldest-order min
ABC, 3 orders/tour	8	242.8	34.3
ABC, 5 orders/tour	6	183.2	47.4
Spread, 5 orders/tour	8	245.7	55.1

ABC with k=5 yields **183.2 paid hours/1,000** (95% CI 177.3-189.1), or about **$4,580/1,000**. That is 24.5% less measured labor than k=3 ABC, but oldest-order completion rises by 13.1 minutes. This batch-leading-order metric is not mean individual-order lead time. At compact demand (152.3/hour), eight pickers with k=5 suffice; ABC provides more headroom than Spread at nearly equal paid labor.

Why the recommendation is conditional

Batching: at 20 compact pickers, k=5 puts ABC back ahead: 410.0 vs. 393.6 orders/hour. **Passing:** removing all aisle interference also restores ABC's advantage. **Traffic control:** 76.6% of ABC's compact blocking at 20 pickers is entry-gate waiting. Exclusive return trips and direction changes are conservative dispatch assumptions, not measured site behavior.

Model and verification

Training uses 2.57M orders in an 80/20 order-ID split; affinities use 120,000 training baskets. Of 60,000 sampled held-out orders, 31,816 are eligible in the larger layout. Four storage policies plus a concentration knob use temperature blocks and frozen-last routes. SimPy tracks travel, pick, entry/cell waiting and idle time; base assumptions are 1.2 m/s, 15 s/line and 30 s depot handling. Separate saturated-capacity and Poisson-demand runs prevent demand caps from hiding congestion.

Checks: 2,900 replicated shifts, 290 cells, ten seeds per cell; one-hour warm-up plus eight measured hours, with a two-hour sensitivity. All 320 one-picker checks agree with independent analytic distances/times to numerical precision. Nine test methods include 24 reference-vs-optimized backend comparisons. Time and order-flow conservation pass. Bands reflect replication uncertainty under fixed assumptions.

Limits before using this in an operation

Geometry, storage temperatures and traffic rules are assumed. There are no weights, volumes, quantities, cart capacities, calendar dates or facility demand. Long-tail work, refrigeration, replenishment, breaks, errors and relocation costs are excluded. Order-level holdout does not isolate customers. Collect actual travel/service times, passing and return rules, batch limits, demand and tail-order costs before deploying a policy.

Sources: Instacart Online Grocery Shopping Dataset 2017; pinned public mirror and hashes in results/data_manifest.json. de Koster et al. (2007), EJOR 182:481-501, doi:10.1016/j.ejor.2006.07.009 (framing). Gue et al. (2006), IIE Transactions 38:859-868, doi:10.1080/07408170600809341 (blocking motivation; author manuscript consulted). Full assumptions, references, code and results accompany this brief.